

$$P = \frac{2\pi a}{\omega} = \frac{1}{f} \quad \omega = \frac{v}{r}$$

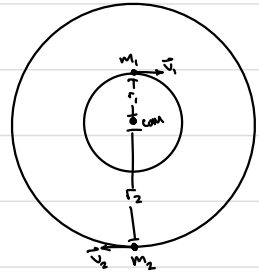
Q2 WTS Newt2 $\rightarrow \vec{F} = \sum \vec{F}_i = m\vec{a}$ $\rightarrow$ Kep3 $\rightarrow P^2 = \frac{4\pi^2}{G(m_1+m_2)} a^3$

a)

$$N=2$$

$$e_1 = e_2 = 0$$

$m_1 = \text{star}$
 $m_2 = \text{planet}$



proof:

WTS Newt2 $\rightarrow$ Kep3 if $N=2, e_1=e_2=0$

$$\sum \vec{F}_i = m\vec{a} \quad \text{for one object}$$

$$|\vec{F}_{\text{grav}}| = |m_1 \vec{a}_c| = |m_1 \omega^2 \vec{r}_1|$$

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

$$\vec{F}_1 = \frac{\mu}{m_1} \vec{r}$$

$$\vec{F}_2 = \frac{\mu}{m_2} \vec{r}$$

drop
if taking mag only

(w2.11)

$$G \frac{m_1 m_2}{r^2} = m_1 \omega^2 r_1$$

$$\frac{G m_2}{r^2} = \omega^2 r_1 \quad \frac{\mu}{m_1} r = m_1 \left(\frac{m_1 m_2}{m_1 + m_2} \right) r$$

$(r_1 + r_2)^2$

$$\frac{G m_2}{r^2} = \omega^2 \frac{m_2}{(m_1 + m_2)} r$$

$$\frac{G(m_1 + m_2)}{r^3} = \omega^2 = \left(\frac{2\pi}{P} \right)^2 = \frac{4\pi^2}{P^2}$$

$$\therefore \boxed{P^2 = \frac{4\pi^2}{G(m_1 + m_2)} r^3}$$

QED

$$b) \text{ Kep3 } P^2 = \frac{4\pi^2}{G(m_1 + m_2)} r^3$$

$$\text{for } m_1 \rightarrow r = \frac{m_1 r_1}{\mu} = \frac{\cancel{m_1}}{\frac{\cancel{m_1} m_2}{(m_1 + m_2)}} r_1 = \frac{m_1 + m_2}{m_2} r_1$$

$$\hookrightarrow P^2 = \frac{4\pi^2}{G} \frac{1}{(m_1 + m_2)} \left(\frac{m_1 + m_2}{m_2} r_1 \right)^3$$

$$P^2 = \frac{4\pi^2}{G} \frac{(m_1 + m_2)^2}{m_2^3} r_1^3$$

$$P^2 = \frac{4\pi^2 r_1^3}{G} \underbrace{\frac{(m_1 + m_2)^2}{m_2^3}}_{\mu^{-1}}$$

(not same μ as earlier)

$$\boxed{P^2 = \frac{4\pi^2 r_1^3}{G \mu}}$$

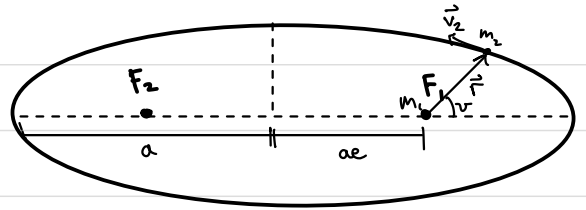
$$, \mu = \frac{m_2^3}{(m_1 + m_2)^2}$$

QED

$$c) \quad m_2 \ll m_1, \quad \text{Kep2} \rightarrow \frac{dA}{dt} = \frac{L}{2m_2} \quad \left(\frac{dA}{dt} = \frac{L}{2\mu} \right)$$

$$e_2 \neq 0$$

proof: WTS Kep3 from Kep2



$$\text{Kep2} \quad \frac{dA}{dt} = \frac{L}{2\mu}$$

$$\text{if } m_2 \ll m_1 \rightarrow \mu = \frac{m_1 m_2}{m_1 + m_2} \approx \frac{m_1 m_2}{m_1} \approx m_2$$

$$\text{then} \quad \frac{dA}{dt} = \frac{L}{2m_2}$$

integ over one P
(sweep over entire A in that time)

$$\int_0^A dA = \int_0^P \frac{L}{2m_2} dt$$

$$A = \frac{L}{2m_2} P$$

$$(2m_2 \pi ab)^2 = (L P)^2$$

$$\rightarrow 4\bar{a}^2 m_2^2 a^2 b^2 = L^2 P^2 \quad \text{generally}$$

@ pericenter ("closest to center") $\rightarrow$ perihelion $\therefore v=0$ & $r=r_p$

$$\hookrightarrow 4\bar{a}^2 m_2^2 a^2 [a^2(1-e^2)] = r_p^2 Gm_1 m_2^2 (1+e \cos v) P^2$$

$$r_p = a - ae = a(1-e)$$

$$4\bar{a}^2 a^3 (1-e^2) = a(1-e) Gm_1 (1+e) P^2$$

$$(1-e) \cdot (1+e) = (1-e^2)$$

$$4\bar{a}^2 a^3 = Gm_1 P^2$$

$$\therefore \boxed{P^2 = \frac{4\bar{a}^2}{Gm_1} a^3} \quad \checkmark \text{ Kep3 when } m_2 \ll m_1$$

QED

$$A = \pi ab$$

$$b^2 = a^2(1-e^2)$$

$$r(v) = \frac{L^2}{Gm_1 m_2^2 (1+e \cos v)}$$

$$v=0 \quad (\text{perihelion})$$

$$v=\pi \quad (\text{aphelion})$$

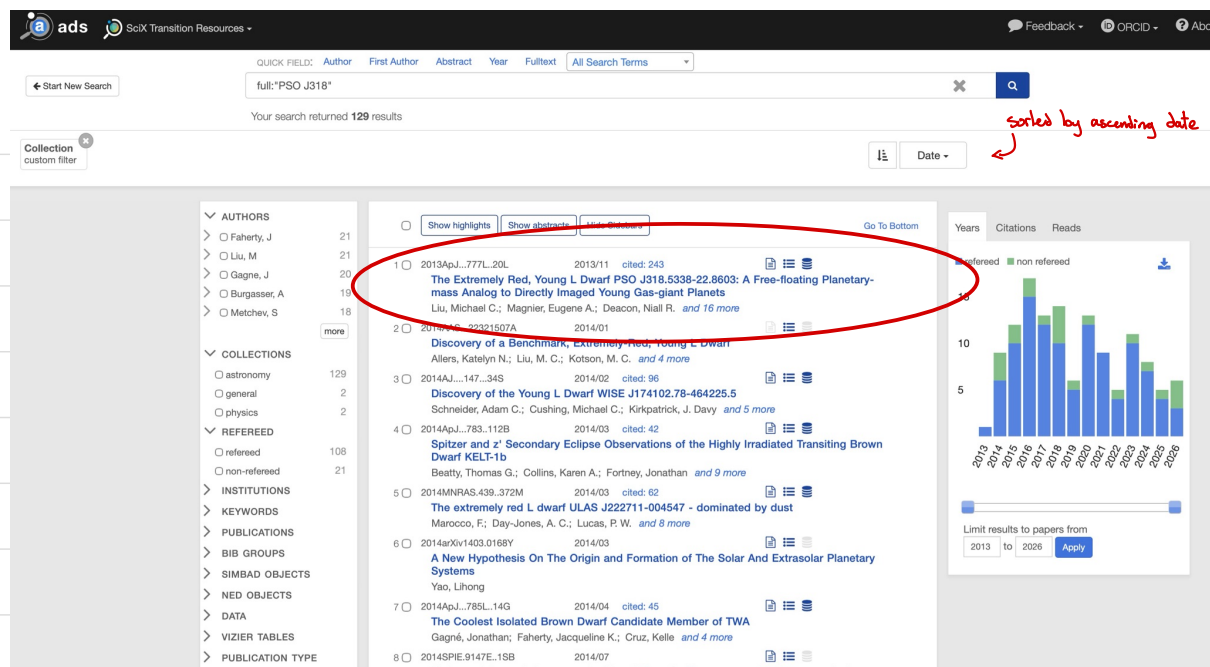
$$L^2 = r Gm_1 m_2^2 (1+e \cos v)$$

if $v=0$

$$\text{then } \cos v = 1$$

$$\hookrightarrow L^2 = r Gm_1 m_2^2 (1+e)$$

Q3



The Extremely Red, Young L Dwarf PSO J318.5338-22.8603: A Free-floating Planetary-mass Analog to Directly Imaged Young Gas-giant Planets

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Liu, Michael C. ; Magnier, Eugene A. ; Deacon, Niall R. ; Allers, Katelyn N. ; Dupuy, Trent J. ; Kotson, Michael C. ; Aller, Kimberly M. ; Burgett, W. S. ; Chambers, K. C. ; Draper, P. W. ; Hodapp, K. W. ; Jedicke, R. ; Kaiser, N. ; Kudritzki, R.-P. ; Metcalfe, N. ; Morgan, J. S. ; Price, P. A. ; Tonry, J. L. ; Wainscoat, R. J.

[1]Liu, M. C., “The Extremely Red, Young L Dwarf PSO J318.5338-22.8603: A Free-floating Planetary-mass Analog to Directly Imaged Young Gas-giant Planets”, *The Astrophysical Journal*, vol. 777, no. 2, Art. no. L20, IOP, 2013. doi:10.1088/2041-8205/777/2/L20.

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Planet Name	Host Name	Default Parameter Set	Number of Stars	Number of Planets	Discovery Method	Discovery Year	Discovery Facility	Solution Type	Controversial Flag	Planetary Parameter Reference	Orbital Period [days]	Orbit Semi-Major Axis [au]	Planet Radius [Earth Radius]	Planet Radius [Jupiter Radius]	Planet Mass or Mass* $\sin(i)$ [Earth Mass]	Planet Mass or Mass* $\sin(i)$ [Jupiter Mass]			
✓ M62H b	1	M62H	1	1	Pulsar Timing	2004	South African Rac	Published Confirmed	0	Vieeschow et al. 2004	0.132935028±0.000000006				>785	>2.47			
✓ PSR B0329+54 b	2	PSR B0329+54	1	1	Pulsar Timing	2017	Multiple Facilities	Published Confirmed	0	Starovoi & Rodin 2017	10140±11	10.26±0.07			1.97±0.19	0.00620±0.0004			
✓ PSR B1257+12 b	3	PSR B1257+12	1	1	Pulsar Timing	1994	Arecibo Observat	Published Confirmed	0	Konacki and Wolzcczan 2003	25.262±0.003	0.19			0.020±0.002	6.3e-05±6.3e-06			
✓ PSR B1257+12 c	4	PSR B1257+12	0	1	Pulsar Timing	1994	Arecibo Observat	Published Confirmed	0	Wolzcczan 1994	25.34±0.05	0.19			0.015	4.7e-05			
✓ PSR B1257+12 d	5	PSR B1257+12	0	1	Pulsar Timing	1992	Arecibo Observat	Published Confirmed	0	Wolzcczan et al. 1992	66.563±0.009	0.36			3.4	0.011			
✓ PSR B1257+12 e	6	PSR B1257+12	0	1	Pulsar Timing	1992	Arecibo Observat	Published Confirmed	0	Wolzcczan 1994	66.536±0.001	0.36			3.4	0.011			
✓ PSR B1257+12 f	7	PSR B1257+12	1	1	Pulsar Timing	1992	Arecibo Observat	Published Confirmed	0	Konacki and Wolzcczan 2003	66.5419±0.0001	0.36			4.3±0.2	0.014±0.001			
✓ PSR B1257+12 g	8	PSR B1257+12	0	1	Pulsar Timing	1992	Arecibo Observat	Published Confirmed	0	Wolzcczan et al. 1992	98.234±0.021	0.47			2.8	0.0068			
✓ PSR B1257+12 h	9	PSR B1257+12	0	1	Pulsar Timing	1992	Arecibo Observat	Published Confirmed	0	Konacki and Wolzcczan 2003	98.214±0.0002	0.46			3.9±0.2	0.012±0.001			
✓ PSR B1257+12 i	10	PSR B1257+12	0	1	Pulsar Timing	1992	Arecibo Observat	Published Confirmed	0	Wolzcczan 1994	98.223±0.002	0.47			2.8	0.0068			
✓ PSR B1620-26 b	11	PSR B1620-26	1	2	Pulsar Timing	2003	Hubble Space Tel	Published Confirmed	0	Sigurdsson et al. 2003		23			795±318	2.5±1.0			
✓ PSR J1719-1438 b	12	PSR J1719-1438	1	1	Pulsar Timing	2011	Parkes Observato	Published Confirmed	0	Balles et al. 2011	0.060706293±0.000000002	0.0044			>382.80	>1.20			
✓ PSR J2322-2650 b	13	PSR J2322-2650	1	1	Pulsar Timing	2017	Parkes Observato	Published Confirmed	0	Spiewak et al. 2018	0.322963997±0.000000006	0.0102			252.6±0.1	0.7949±0.0002			

Input "Pulsar Timing" in the discovery method field.

Only the exoplanets discovered via this method show up.

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∴ 8 exoplanets discovered via pulsar timing